
EM 504: Materials for Energy Applications

Course Code	EM 504
Course Name	Materials for Energy Applications
L-T-P-C	3-1-0-4
Prerequisites	None
Intended for	3rd and 4th Year UG/PG
Approval	8th Senate (Oct 2015)

Course Contents

- **Module I**

Creep resistance materials: principles of creep deformation, mathematical modeling and design application of creep resistance materials in power plants. Catalysis; Wear resistance materials: types of wear, mechanism of erosion and cavitation wear, tribo-corrosion, application of wear resistance materials in power plants, Fatigue types, types and protection. Life prediction of turbine materials. [15 Lectures]

- **Module II**

Phase diagram, Conductors, transformer materials, Semiconductor physics and devices, Fundamentals and energy applications for Magnetic and Dielectric materials. [20 Lectures]

- **Module III**

Energy efficient structure/building materials. [7 Lectures]

Text Books

1. Hans Berns , Werner Theisen, Gillian Scheibelein, **Ferrous Materials**, Springer, 2008.
2. A. J. Moulson and J. M. Herbert, **Electroceramics: Materials, Properties and Applications**, 2nd Edition, Wiley, 2003.
3. M.Hall, **Materials for Energy Efficiency and Thermal Comfort in Buildings**, Elsevier Press, 2010.

References

1. Oliver Gutfleisch , Matthew A. Willard , Ekkes Bruck , Christina H. Chen , S. G. Sankar , and J. Ping Liu, **Magnetic Materials and Devices for the 21st Century: Stronger, Lighter, and More Energy Efficient**, *Adv. Mater.*, 23, 821-842 (2011)
2. L.H.Lewis and F.J. Villacorta, **Perspectives on Permanent Magnetic Materials for Energy Conversion and Power Generation**, *Metallurgical and Mater. Trans. A*, 44A, S1-S20 (2013)
3. S.L. Swartz, **Topics in Electronic Ceramics**, *IEEE Trans. Elect. Insulation*, 25 [5] 935 (1990)